Geometric Evaporation and Hierarchical Information Reduction in Black Holes:

The $\mathrm{K}=12 \to \mathrm{K}=0$ Lattice Phase Transition in the SSM Framework

Raghu Kulkarni*

SSMTheory Group, IDrive Inc., Calabasas, CA 91302, USA

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Abstract

We develop a general theory of black-hole evaporation and information dynamics in the Selection-Stitch Model (SSM): a discrete $\mathrm{K}=12$ FCC tensor-network vacuum [1, 2] with $[[192, 130, 3]]$ CSS stabilizer code [3] and $G = a^2/(8\ln 2)$ fixed by Bekenstein–Hawking entropy matching [4]. A black hole is a *topological vacancy* where gravitational compression has driven the lattice past the metric wall $r_{\min} = L_0/\sqrt{3}$ [1], shattering all entanglement bonds and dropping the local coordination from $\mathrm{K}=12$ to $\mathrm{K}=0$. Treating the horizon as a 2D phase boundary with surface tension $\sigma = \hbar c/(4L_0^3)$ and applying absolute quantum rate theory to boundary re-stitching, we derive a recession velocity $\dot{R} = -(c/2)(L_0/R_H)$ and a geometric lifetime $\tau_{\mathrm{Geo}} = 4G^2M^2/(c^5 L_0) \propto M^2$, holding for all black holes. Peierls locking at correlation length $L_{\mathrm{corr}} \sim 1$ fm suppresses this channel exponentially for $R_H \gg L_{\mathrm{corr}}$, so stellar-mass and supermassive black holes evaporate only through the standard Hawking channel; the geometric channel is observable only for primordial black holes (PBHs) with $R_H \lesssim L_{\mathrm{corr}}$. A 10^{15} g PBH evaporates in 0.27 ms, resolving the long-standing gamma-ray constraint that requires $\tau_{\mathrm{Hawk}} \propto M^3$ relics today. We frame the information question hierarchically: information in the SSM exists at four levels (bond, plaquette, cuboctahedral cell, global stabilizer pattern). Across the $\mathrm{K}=12 \to \mathrm{K}=0$ boundary the hierarchy itself does not survive: it *reduces* to the lowest level. Bond-level information is preserved as the bit count of severed bonds, emerging as thermal Hawking radiation and reproducing $S_{\mathrm{BH}} = A/(4G)$; higher-level structure collapses into that thermal flux. Hawking's exact-thermal result is recovered, and the absence of Page-curve recovery of higher-level structure is a falsifiable prediction.

1 Introduction

The Hawking radiation mechanism [5] predicts a thermal emission rate that gives a black-hole lifetime $\tau_{\mathrm{Hawk}} \propto M^3$, derived from quantum field theory on a fixed curved background. This calculation suffers from the trans-Planckian problem: the outgoing radiation modes are traced to field fluctuations with sub-Planckian wavelengths, where the spacetime continuum approximation must break down. The Selection-Stitch Model (SSM) framework [1–4] replaces the spacetime continuum with a Face-Centered Cubic (FCC) lattice of coordination number $\mathrm{K}=12$ and lattice spacing $a \approx 2.355\,\ell_P$. The trans-Planckian assumption becomes a hard ultraviolet cutoff at $|\vec{k}| \lesssim \pi/a$ [4], and Hawking's calculation must be re-examined given the underlying discrete dynamics.

This Letter develops a general theory of black-hole evaporation and information dynamics in the SSM framework, applicable to all black holes regardless of mass. The two main results—a geometric evaporation channel and a hierarchical reduction of information at the horizon—hold

*raghu@idrive.com

universally. Whether they are observationally distinguishable from standard Hawking physics depends on a single dimensionless ratio: $R_H/L_{\rm corr}$, where $L_{\rm corr} \sim 1$ fm is the QCD-set structural correlation length of the discrete vacuum. For $R_H \gg L_{\rm corr}$ (stellar-mass and supermassive black holes), Peierls locking suppresses the geometric channel exponentially and only the standard Hawking decay survives; the framework reproduces standard astrophysical-BH phenomenology. For $R_H \lesssim L_{\rm corr}$, the geometric channel dominates, giving observable differences. The latter regime corresponds to primordial black holes (PBHs) near the historically constrained 10^{15} g threshold [6], providing both the observational motivation for the present analysis and a concrete falsifiable target.

Geometric evaporation. A black hole is a topological vacancy where the FCC lattice has been compressed past the metric wall $r_{\rm min} = L_0/\sqrt{3}$ [1], shattering all entanglement bonds and reducing the local coordination to $\mathrm{K}=0$. The horizon is a discrete 2D phase boundary between the $\mathrm{K}=12$ vacuum and the $\mathrm{K}=0$ interior, with calculable surface tension. Boundary re-stitching dynamics, derived from absolute quantum rate theory [7], gives a recession velocity $\dot{R} = -(c/2)(L_0/R_H)$ and a lifetime $\tau_{\rm Geo} = 4G^2M^2/(c^5 L_0)$ holding for any M. For 10^{15} g PBHs this gives 0.27 ms, in contrast to Hawking's prediction of $\sim 10^{20}$ s; for stellar-mass and supermassive black holes, Peierls locking shuts the channel down, and standard Hawking lifetimes survive.

Hierarchical information reduction. We address the information question by distinguishing four levels of information content in the SSM framework: (L1) bond-level—the existence of individual indistinguishable Bell pairs; (L2) plaquette-level—local correlations between the four bonds of a CSS stabilizer; (L3) cell-level—the configuration of the twelve bonds in a cuboctahedral $\mathrm{K}=12$ neighborhood; and (L4) global-level—the pattern of stabilizer eigenvalues defining the logical-qubit state of the $[[192, 130, 3]]$ code. This applies to every black hole, regardless of size: any $\mathrm{K}=12 \to \mathrm{K}=0$ horizon transition reduces the information hierarchy to its lowest level. Bond-level information (L1) is preserved as the bit count of severed bonds, emerging as exactly thermal Hawking radiation; higher-level structure (L2–L4)—the combinatorial pattern of which Bell pairs occupied which edges—does not propagate across the boundary as structure but collapses into the L1 thermal flux. The Bekenstein–Hawking entropy $S = A/(4G)$ is recovered as the L1 bit count of the horizon. Hawking's exact-thermal radiation calculation is reproduced naturally: the emitted quanta carry maximal entropy per energy, which is precisely L1 information. The absence of Page-curve-style recovery of higher-level structure is a sharp falsifiable prediction holding for every black hole, but observable in evaporation products only for PBHs near the 10^{15} g threshold where geometric evaporation is fast enough to study.

2 Black Holes as Topological Vacancies

2.1 The metric wall and lattice shattering

The FCC vacuum has $\mathrm{K}=12$ nearest neighbors per site at bond length $L_0 = a/\sqrt{2}$ [1]. The space between four mutually touching FCC sites (a tetrahedral void) has inscribed-sphere radius

$$r_{\rm min} = \frac{L_0}{\sqrt{3}}. \tag{1}$$

This is the minimum internode distance compatible with FCC topology: below $r_{\rm min}$, the tetrahedral voids collapse and the lattice cannot maintain its $\mathrm{K}=12$ connectivity. The matter paper [1] establishes $r_{\rm min}$ as the absolute exclusion limit of the discrete vacuum; gravitational compression past this wall is geometrically impossible within the $\mathrm{K}=12$ phase.

2.2 The K = 0 topological vacancy

Definition 1 (Topological vacancy)**.** *A black hole in the SSM framework is a connected region $\mathcal{V} \subset \mathbb{R}^3$ in which the local gravitational compression has forced the internode spacing below $r_{\min}$. Within $\mathcal{V}$, all tetrahedral voids have collapsed and all entanglement bonds passing through $\mathcal{V}$ have been severed. The local coordination number drops from $K = 12$ (intact vacuum) to $K = 0$ (vacancy interior). The boundary $\partial\mathcal{V}$ between the intact lattice and the vacancy is the event horizon.*

The $\mathrm{K} = 12 \to \mathrm{K} = 0$ transition is a discrete topology change: not a smooth deformation but a destruction of network connectivity. The interior of $\mathcal{V}$ contains no surviving lattice degrees of freedom and is causally isolated from the exterior. The boundary $\partial\mathcal{V}$ consists of *dangling Bell-pair halves*—bonds whose interior partners have been destroyed and which radiate into the exterior with characteristic energy $E_{\text{bond}} = \hbar c/(4L_0)$ [4].

2.3 Recovery of standard thermodynamic black-hole properties

The topological-vacancy model reproduces the three central thermodynamic properties of black holes from the underlying CSS-code structure.

Proposition 2 (Bekenstein–Hawking entropy)**.** *The entropy of a topological vacancy of horizon area A is*

$$S = \frac{A}{4G}, \qquad G = \frac{a^2}{8\ln 2}. \tag{2}$$

Proof. Each removed Ovoid X-stabilizer of the CSS code [3] on the horizon contributes $\Delta S = \ln 2$ to the entanglement entropy between interior and exterior, and occupies area $A_{\text{plaq}} = L_0^2 = a^2/2$ in the 2D triad-sheet plaquette structure [4]. Bekenstein–Hawking entropy matching at the level of a single stabilizer gives $G = A_{\text{plaq}}/(4\ln 2) = a^2/(8\ln 2)$ (see [4], eq. 12). For a horizon of area A comprising $N = A/L_0^2$ severed stabilizers, $S = N\ln 2 = A\ln 2/L_0^2 = A/(4G)$ via the algebraic identity $L_0^2 = 4G\ln 2$. □

Proposition 3 (Schwarzschild radius)**.** *The boundary of the topological vacancy coincides with the Schwarzschild radius $R_H = 2GM/c^2$.*

Proof. The vacancy nucleates at the center of the gravitating mass, where the energy density of gravitational self-compression first reaches the metric-wall density. Once nucleated, the vacancy propagates outward through every shell whose enclosed mass $M(r)$ satisfies the trapped-surface condition $2GM(r)/(rc^2) \geq 1$. The outermost such shell is precisely $r = 2GM/c^2 = R_H$. □

Proposition 4 (Hawking temperature)**.** *The dangling-bond emission spectrum from the horizon is thermal with temperature $T_H = \hbar c^3/(8\pi GMk_B)$.*

Proof. The dangling Bell-pair halves on $\partial\mathcal{V}$ are in detailed balance with the intact $\mathrm{K} = 12$ lattice exterior at the local Tolman-rescaled vacuum temperature. The Unruh relation $T = \hbar\kappa/(2\pi k_B c)$, with surface gravity $\kappa = c^4/(4GM)$ at the Schwarzschild horizon, gives the stated T_H. □

The SSM topological-vacancy model is therefore consistent with the three thermodynamic black-hole laws of General Relativity, with all three numerical coefficients fixed by the single CSS-code parameter $G = a^2/(8\ln 2)$.

3 Hierarchical Information Reduction

We now address the central conceptual question raised by black-hole evaporation: what happens to the quantum information thrown into the black hole? Our answer is that information itself is conserved in its lowest form, while the *hierarchical structure* of information is not. This requires distinguishing four *levels* of information content in the SSM framework, all four of which are present in the $\mathrm{K}=12$ vacuum but which transform differently across the $\mathrm{K}=12 \to \mathrm{K}=0$ phase boundary: the hierarchy collapses, leaving only the L1 thermal flux that emerges as Hawking radiation. This is the same pattern that governs every irreversible thermodynamic process—a burning library preserves its atoms (lowest level) while losing the organized hierarchy of ink, words, books, and catalog—and the black-hole case is a clean realization of it.

3.1 Four levels of information

In the SSM framework, the vacuum is a tensor network: a specific decoration of the FCC lattice with Bell-pair bonds on each of the $K = 12$ edges per site, organized into a $[[192, 130, 3]]$ CSS stabilizer code [3]. The information content of this state is hierarchical, with each higher level encoding more complex combinatorial structure built from lower-level constituents.

- *Level 1 (bond-level)*: the existence of an individual Bell pair on a given edge. Each bond carries $\ln 2$ of entanglement entropy. Bell pairs are intrinsically indistinguishable: every $|\Phi^+\rangle_{vv'}$ is identical to every other, with no individual label. L1 information is the cardinality of the set of present bonds—a counting statistic, not a structural one.
- *Level 2 (plaquette-level)*: the local correlations between the four bonds of a CSS stabilizer plaquette. Each weight-4 stabilizer $A_p = X_{e_1} X_{e_2} X_{e_3} X_{e_4}$ enforces a constraint on the joint state of four neighboring edges; its eigenvalue (± 1) is one bit of L2 structural information. L2 information distinguishes between configurations with the same L1 bond count but different local correlation patterns.
- *Level 3 (cell-level)*: the configuration of the twelve bonds in the cuboctahedral $\mathrm{K}=12$ neighborhood of a single FCC site. Each cell encodes the local correlation pattern among its twelve edges, plus the relations between adjacent stabilizer plaquettes. This is the natural unit of *geometric* information in the framework—the discrete analog of a curvature or strain measurement at a point.
- *Level 4 (global-level)*: the pattern of stabilizer eigenvalues across the entire $[[192, 130, 3]]$ CSS code. With $n = 192$ physical qubits and $k = 130$ logical qubits, the L4 information content is up to 2^{130} logical states distinguished by the global eigenvalue pattern of the $n - k = 62$ stabilizers. This is the most complex form of information: the macroscopic quantum-information state of the encoded computational sector.

Each level is built from constituents of the previous level, but the higher-level information is not reducible to enumeration of the lower-level constituents: it lives in their relational structure. The library analogy is direct: ink (L1), letters (L2), words and sentences (L3), and the meaningful text and the catalog as a whole (L4) are increasingly complex structural levels built from progressively richer relations among the simpler ingredients.

3.2 Bond-level information is preserved as thermal Hawking radiation

When the $\mathrm{K}=12$ lattice shatters at the metric wall, every Bell pair passing through the vacancy region is severed. The number of severed bonds is the bit count of bond-level information present in the interior, plus the bit count of bond-level information on the boundary itself.

The bit count is preserved. Each severed bond becomes a dangling Bell-pair half on $\partial\mathcal{V}$, which subsequently radiates one quantum of thermal Hawking radiation into the exterior (Proposition 4). The total number of thermal quanta emitted over the lifetime of the black hole equals the total number of bonds severed during its formation and existence, which equals the horizon's L1 information content:

$$N_{\text{thermal quanta}} = N_{\text{severed bonds}} = \frac{A_{\text{horizon}}}{L_0^2} = \frac{S_{\text{BH}}}{\ln 2} = \frac{A_{\text{horizon}}}{4G\ln 2}. \tag{3}$$

The thermal nature of the radiation is the signature of L1-only information. A thermal distribution at temperature T_H has maximum entropy for given total energy: it carries one bit per emitted quantum (the existence of that quantum) and nothing more. This is exactly L1 information, propagating outward—no structural information at L2 or higher is encoded in the spectrum. Hawking's exact-thermal calculation [5] is recovered as the L1 information channel of the SSM framework.

The Bekenstein–Hawking entropy is therefore the bit count of L1 information stored at the horizon:

$$S_{\text{BH}} = N_{\text{severed}} \ln 2 = \frac{A}{4G}, \tag{4}$$

counting one $\ln 2$ per severed bond. This is the L1 bit count of the horizon, neither more nor less.

3.3 Higher-level structure reduces to the L1 thermal flux

L2, L3, and L4 information do *not* propagate across the $\mathrm{K}=12 \to \mathrm{K}=0$ phase boundary as structure: they reduce to the L1 thermal flux, becoming inaccessible as higher-level information. The bit count is preserved (Section 3.2); the hierarchical structure is not.

L2 reduces. The CSS stabilizer eigenvalues are joint properties of four edges sharing a plaquette. When all four edges are severed simultaneously, the eigenvalue ceases to be a well-defined quantity: the operator $A_p = X_{e_1}X_{e_2}X_{e_3}X_{e_4}$ acts on a Hilbert space whose dimension has changed, and the pre-severance eigenvalue is not recoverable from the post-severance dangling-bond ensemble. The dangling halves are statistically independent thermal states whose correlations with the original stabilizer eigenvalue are washed into the L1 thermal flux.

L3 reduces. The cuboctahedral cell structure—the geometric arrangement of twelve bonds around a single FCC site—requires the FCC site itself to exist as a coordination center. When the site is shattered (its inscribed-sphere radius forced below $r_{\min}$), the cell ceases to exist as a structural unit. The twelve bonds it formerly organized are scattered into the boundary dangling-bond ensemble; their identities as constituents of a particular cell are not retained.

L4 reduces. The global logical-qubit state of the $[[192, 130, 3]]$ code is determined by the joint eigenvalues of all 62 stabilizers [3]. When a macroscopic fraction of stabilizers (everything in the vacancy region) is severed, the global logical state is no longer defined: its information content reduces to the L1 bit count of the severed bonds. The post-evaporation lattice may recrystallize into a new $\mathrm{K}=12$ configuration (Section 3.5), but the new configuration's L4 state is statistically independent of the predecessor—a fresh draw from the ensemble of possible vacuum configurations, not a unitary transformation of the original.

Higher-level information reduces to L1 because indistinguishable constituents cannot maintain relational structure once their relations are broken. The same indistinguishability that makes L1 a pure counting statistic makes L2, L3, and L4 information dependent on bond-by-bond relationships that do not survive bond severance. What survives is the count—and the count is exactly what emerges as thermal Hawking radiation. The library analogy is precise: burning an ordered library preserves all the atoms (L1 of that system) while collapsing the entire hierarchy of letters, words, books, and catalog (L2–L4 of that system) into thermal motion. The atoms in the smoke and ash are exactly the atoms that were in the books; what is gone is the organization. The $K = 12 \to K = 0$ transition is the same phenomenon at the level of vacuum bonds.

3.4 Unitarity at each level

The hierarchical structure refines the meaning of "unitarity" in the framework:

L1 unitarity is fundamental and globally preserved. The total bit count of bond-level information is conserved across the $K = 12 \to K = 0$ transition: every severed bond emerges as one thermal quantum, with no L1 bits created or destroyed. By the mass-energy-information equivalence established in the MEI paper [2], L1 conservation is not merely analogous to energy conservation but identical to it—the bit count of severed bonds and the energy carried away as thermal Hawking radiation are two expressions of the same conserved quantity in the SSM framework. This conservation holds exactly even at the phase boundary. The Banks–Susskind–Peskin argument [10] that local non-unitarity violates energy conservation or locality does not apply: L1 unitarity is preserved locally, energy is conserved by the MEI equivalence, and the dynamics is local within the $K = 12$ phase.

L2–L4 unitarity is emergent within the $\mathbf{K = 12}$ phase. The CSS-stabilizer-code dynamics [3, 4] preserves L2, L3, and L4 information by construction within the $K = 12$ phase: stabilizer measurements and code operations are unitary maps on the protected logical Hilbert space. All standard physics—atomic, nuclear, particle, quantum-information experiments—lives within this regime, and sees full unitarity at all four levels. The strict unitarity assumed in standard quantum mechanics is, in the SSM framework, the L4 unitarity of the $K = 12$ phase.

Higher-level unitarity does not extend across phase boundaries. The $K = 12 \to K = 0$ transition is a non-equilibrium phase transition where the CSS-code Hilbert space itself changes dimension. Unitarity at L2–L4, defined as a norm-preserving map on a fixed Hilbert space, simply does not apply across this boundary: the hierarchy on which L2–L4 are defined has collapsed. What remains is L1 unitarity, which holds globally and which is what conventional energy/probability conservation requires. The standard "information paradox" arises only from assuming strict L4 unitarity in a regime where the lattice phase boundary makes that assumption inapplicable.

3.5 Phase-boundary processes: black holes and the Big Bang

The SSM framework predicts that two physical processes reduce the hierarchical information structure to its lowest level:

Black-hole formation and evaporation ($\mathbf{K = 12 \to K = 0 \to K = 12}$). Gravitational collapse drives the $K = 12$ vacuum into $K = 0$ vacancy, reducing interior L2–L4 information to the L1 bit count of severed bonds. Subsequent geometric evaporation (Section 4) re-stitches $K = 12$ lattice across the former vacancy region, producing a new specific configuration whose L4 state is statistically independent of the predecessor. The radiation carries away the L1 bit count; the L2–L4 organization of the infallen matter is reduced to that thermal flux, and a new L4 sector is freshly crystallized in the re-stitched vacuum.

Big Bang crystallization (phased $\mathbf{K=0 \to K=6 \to K=4 \to K=12}$). The matter paper [1] establishes that the FCC vacuum does not crystallize from disorder in a single transition: the $K=12$ vacuum emerges through an ordered sequence of intermediate phases. From the initial high-temperature $K=0$ disordered phase, two-dimensional triangular sheets crystallize first (giving in-plane coordination $K=6$), with occasional out-of-plane lifts producing transitional $K=4$ sites, and three such sheets eventually register into the FCC stacking that yields full cuboctahedral $K=12$ coordination. The Big Bang in this picture is specifically the final $K=4 \to K=12$ sheet-projection event that establishes the asymptotic vacuum in which all subsequent physics operates.

The hierarchical information content emerges in step with the geometric phases:

- L1 (bond existence) appears as soon as the first 2D sheet forms ($K=6$).
- L2 (plaquette correlations) becomes well-defined within each sheet, with full four-edge correlations only present at $K=12$ once cross-sheet bonds register.
- L3 (cuboctahedral cell) requires the $K=12$ coordination and only exists in the final phase.
- L4 (global $[[192, 130, 3]]$ logical state) stabilizes after the full FCC registration, when the complete CSS-stabilizer structure first exists.

Each level of information is therefore created at a different stage of cosmological crystallization, with no precursor in earlier phases. The L4 state of the present cosmic vacuum was selected stochastically during the final $K=4 \to K=12$ transition; it is not unitarily related to any pre-Big-Bang quantum state.

Both processes are non-equilibrium phase transitions in the lattice coordination structure. Neither is described by Hamiltonian evolution within a fixed Hilbert space; both lie outside the regime of validity of standard L4 unitarity.

3.6 Response to standard objections

Three standard objections to non-unitary BH-evaporation scenarios deserve direct response within the hierarchical-reduction framework.

Banks–Susskind–Peskin [10]. The argument is that local non-unitary evolution generically violates energy conservation or locality. Resolution: L1 unitarity is preserved everywhere, including at the phase boundary; the bit count of severed bonds equals the bit count of emitted thermal quanta, and this counting is local at every step. The reduction of L2–L4 structure to the L1 thermal flux is not local non-unitarity in the BSP sense—it is a topology change in the underlying Hilbert space, accompanied by exact L1 conservation. Energy conservation and locality at L1 are preserved, so the BSP argument does not apply.

Page curve [11]. The argument is that if BH evaporation is unitary at *all* levels, the entanglement entropy of the emitted radiation must follow a specific curve (rising then falling). Recent replica-wormhole derivations [12, 13] reproduce this curve in toy models. Resolution: the Page curve assumes L4 unitarity at every step. In the SSM framework, L4 structure does not survive the $K=12 \to K=0$ reduction, and the radiation entropy rises monotonically without ever falling. The replica-wormhole derivations rely on a smooth gravitational path integral across the horizon, which does not exist in the SSM framework (the horizon is a discrete phase boundary, not a smooth Riemannian region). The Page curve is not a prediction of our framework, and its experimental absence would be a confirmation.

AdS/CFT and holography. If a black hole in AdS is dual to a unitary boundary CFT, L4 unitarity holds by construction. Resolution: AdS/CFT applies in a specific setting (asymptotically AdS spacetimes), but the SSM framework is asymptotically flat with a discrete K = 12 FCC vacuum. The AdS/CFT duality does not extend to our setting without modification, and the SSM vacuum does not have an obvious holographic CFT description. The framework is not in conflict with AdS/CFT results; it lies outside their domain of applicability.

3.7 Summary of the hierarchical picture

Level	Information content	Carrier	Survives as structure?
L1 (bond)	Existence of one Bell pair; 1 bit ($\ln 2$) per bond	Severed-bond thermal quantum	**Yes** (as Hawking radn)
L2 (plaquette)	CSS stabilizer eigenvalue; weight-4 correlation	Stabilizer plaquette (A_p)	No (reduces to L1)
L3 (cell)	Cuboctahedral K = 12 neighborhood configuration	Local 12-bond pattern	No (reduces to L1)
L4 (global)	$[[192, 130, 3]]$ logical-qubit state	Joint stabilizer pattern	No (reduces to L1)

Table 1: Four levels of information in the SSM framework and their fate across the K = 12 → K = 0 phase boundary. L1 information is preserved and emerges as thermal Hawking radiation with total bit count $A/(4G \ln 2)$, recovering the Bekenstein–Hawking entropy. L2–L4 do not survive as structural information: they reduce to the L1 thermal flux. The total information is conserved at the lowest level; the hierarchy is not.

The hierarchical picture summarized in Table 1 reconciles three apparently conflicting requirements: (i) Hawking's exact-thermal radiation calculation, recovered as the L1 information channel; (ii) the Bekenstein–Hawking entropy $S = A/(4G)$, which is the L1 bit count of the horizon; and (iii) the experimentally absent Page-curve signature, correctly predicted to be absent because L2–L4 structure reduces to L1 and is not recoverable from the thermal flux. Hawking's original 1975 conclusion [5] that the radiation is exactly thermal is precisely the statement that only L1 emerges from the horizon. The framework's substantive position is therefore: *total information is preserved at the lowest level, but the information hierarchy itself collapses.* This is the same pattern that governs every irreversible thermodynamic reduction in nature.

4 Geometric Evaporation Dynamics

4.1 Horizon surface tension

Theorem 5 (Discrete surface tension)**.** *The surface tension of the $K = 12/K = 0$ phase boundary is*

$$\sigma = \frac{\hbar c}{4L_0^3}. \tag{5}$$

Proof. We do not require a microscopic model of how a cuboctahedral cell dissolves: the order in which individual bonds sever, and any intermediate excited configurations of the cell, are unknown. What is fixed by the framework is the total bond content and the total energy released per unit horizon area.

The horizon $\partial\mathcal{V}$ is a 2D surface intersecting the lattice. By the Bekenstein–Hawking matching (Proposition 2), the entropy per unit horizon area is $\ln 2/L_0^2 = 1/(4G)$, corresponding to one removed CSS stabilizer per area L_0^2 of the 2D triad-sheet plaquette structure [4]. Each removed unit dissolves into Bell-pair halves whose characteristic energy is $E_{\text{bond}} = \hbar c/(4L_0)$ [4]. The total surface energy per unit horizon area is therefore

$$\sigma = \frac{E_{\text{bond}}}{L_0^2} = \frac{\hbar c}{4L_0^3}, \tag{6}$$

independent of the microscopic dissolution sequence. □

4.2 Boundary re-stitching dynamics

The K = 0 vacancy is unstable: the surrounding intact lattice exerts mechanical pressure $P = 2\sigma/R_H$ inward on the boundary, driving spontaneous re-stitching of K = 12 lattice across $\partial\mathcal{V}$. The dynamics is set by the quantum transition rate for a single K = 0 boundary node to acquire its $K = 12$ coordination through re-bonding with neighbors.

Theorem 6 (Recession velocity)**.** *The continuum limit of the discrete quantum transition rate for boundary re-stitching gives a horizon recession velocity*

$$\dot{R} = -\frac{cL_0}{2R_H}. \tag{7}$$

Proof. By absolute quantum rate theory [7], the drift velocity of a phase boundary in response to free-energy difference W per quantum step is $v = (W/\hbar)\,\lambda$, where λ is the elementary step length. In a discrete lattice, the natural step is one bond: $\lambda = L_0$, with corresponding step volume $V_{\text{step}} = L_0^3$. The thermodynamic work done by the boundary pressure $P = 2\sigma/R_H$ over one step volume, with σ given by Theorem 5, is

$$W = P\,L_0^3 = \frac{2\sigma L_0^3}{R_H} = \frac{2L_0^3}{R_H} \cdot \frac{\hbar c}{4L_0^3} = \frac{\hbar c}{2R_H}. \tag{8}$$

The radial recession velocity is therefore $\dot{R} = -WL_0/\hbar = -(c/2)(L_0/R_H)$, with the minus sign indicating inward motion. □

This radial recession is mathematically isomorphic to mean-curvature flow [8] (the Allen–Cahn equation describing antiphase-boundary motion), but derived here from first-principles quantum rate theory on the discrete lattice rather than from a continuum free-energy variational principle.

4.3 Mass-loss rate and lifetime

Combining the Schwarzschild relation $R_H = 2GM/c^2$ (Proposition 3) with the recession velocity (7) of Theorem 6:

$$\dot{M} = \frac{c^2}{2G}\,\dot{R} = -\frac{c^5 L_0}{8G^2 M}. \tag{9}$$

Separating and integrating from initial mass M to $M = 0$:

$$\int_0^M M'\,dM' = \frac{c^5 L_0}{8G^2}\,\tau_{\text{Geo}} \quad\Longrightarrow\quad \tau_{\text{Geo}} = \frac{4G^2}{c^5 L_0} M^2. \tag{10}$$

Using the algebraic identity $L_0 = \sqrt{4G\ln 2}$ from $G = a^2/(8\ln 2)$, and expressing in Planck units $G = \hbar c/m_P^2$, $\ell_P = \hbar/(m_P c)$, $t_P = \ell_P/c$, with $L_0/\ell_P = \sqrt{4\ln 2}$:

$$\boxed{\tau_{\text{Geo}} = \frac{4\ell_P}{L_0}\,t_P\left(\frac{M}{m_P}\right)^2 = \frac{2}{\sqrt{\ln 2}}\,t_P\left(\frac{M}{m_P}\right)^2 \approx 2.40\,t_P\left(\frac{M}{m_P}\right)^2.} \tag{11}$$

This $\tau \propto M^2$ scaling is parametrically faster than Hawking's $\tau \propto M^3$.

For a PBH at the observational threshold $M = 10^{15}$ g $= 10^{12}$ kg, with $M/m_P \approx 4.6 \times 10^{19}$:

$$\tau_{\text{Geo}}(10^{15}\,\text{g}) = 2.40 \times (5.4 \times 10^{-44}\,\text{s}) \times (4.6 \times 10^{19})^2 \approx 0.27\,\text{ms}. \tag{12}$$

Hawking's prediction for the same PBH gives $\tau_{\text{Hawk}} \approx 2700$ Gyr, i.e. ~ 200 times the age of the universe. The geometric channel is faster by a factor $\sim 10^{23}$. PBHs in the SSM framework do not survive into the present era; they evaporate within a fraction of a millisecond of their formation, leaving no detectable gamma-ray signal at the threshold mass.

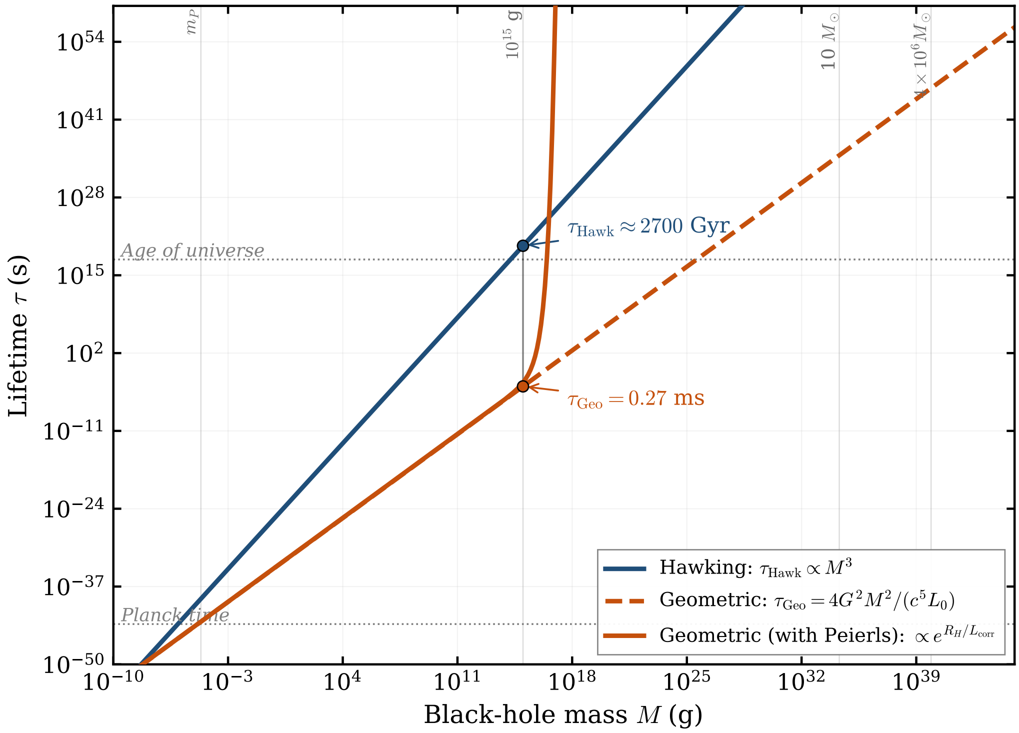

Figure 1: Black-hole lifetime as a function of mass under three scenarios. *Blue:* standard Hawking radiation, $\tau_{\text{Hawk}} \propto M^3$. *Orange dashed:* unobstructed geometric channel, $\tau_{\text{Geo}} = 4G^2M^2/(c^5L_0)$, parametrically faster. *Orange solid:* geometric channel with Peierls suppression $\exp(R_H/L_{\text{corr}})$ active above the QCD correlation length $L_{\text{corr}} \sim 1$ fm. At the observational PBH threshold $M = 10^{15}$ g, the geometric channel gives $\tau_{\text{Geo}} = 0.27$ ms (against Hawking's ~ 2700 Gyr), resolving the gamma-ray constraint. Above $\sim 10^{15}$ g the Peierls factor switches on sharply, locking stellar-mass and supermassive black holes against geometric evaporation; these decay only through the standard Hawking channel.

5 Peierls Locking and Macroscopic Stability

Equation (11) as stated would predict rapid geometric evaporation of all black holes, including stellar-mass and supermassive ones, in conflict with astrophysical observation. This is averted by the Peierls–Nabarro mechanism for discrete domain walls; the resulting sharp suppression of geometric evaporation above $R_H \sim L_{\text{corr}}$ is visible in Fig. 1 as the rapid upturn of the solid orange curve.

5.1 The Peierls suppression

In any discrete lattice, a moving phase boundary cannot glide continuously: it must hop between lattice sites, traversing a periodic potential (the Peierls stress) [9]. For an extended interface of characteristic curvature radius R_H in a polycrystalline matrix with structural correlation length L_{corr}, the boundary mobility is exponentially suppressed:

$$\mu_{\mathrm{eff}} = \mu_0 \exp\left(-\frac{R_H}{L_{\mathrm{corr}}}\right). \tag{13}$$

For microscopic boundaries $R_H \lesssim L_{\mathrm{corr}}$, the exponential is $O(1)$ and the boundary moves freely; for $R_H \gg L_{\mathrm{corr}}$, the exponential is catastrophic and the boundary is locked in place.

5.2 Identification $L_{\mathrm{corr}} \sim 1$ fm

The structural correlation length of the SSM lattice is set by the QCD confinement scale [1]: the natural length over which the FCC-vacuum stabilizer-code structure remains coherent before being disrupted by gauge-field fluctuations is $L_{\mathrm{corr}} \sim 1$ fm.

For the 10^{15} g PBH, $R_H \approx 1.5$ fm, so $R_H/L_{\mathrm{corr}} \approx 1.5$ and the Peierls suppression factor is mild ($e^{-1.5} \approx 0.22$). Geometric evaporation proceeds with only a modest reduction in rate.

For a stellar-mass black hole ($M = 10\,M_\odot$, $R_H \approx 30$ km), $R_H/L_{\mathrm{corr}} \approx 3 \times 10^{19}$, giving a suppression factor of order $\exp(-3 \times 10^{19})$, indistinguishable from zero for any astrophysical purpose. Stellar-mass and supermassive black holes are therefore geometrically stable in this framework, and decay only via the standard Hawking channel.

Black hole	M **(g)**	R_H	**Geometric lifetime**
PBH (Planck mass)	2.2×10^{-5}	3.2×10^{-33} cm	$\sim 2.4\,t_P$
PBH (threshold)	10^{15}	1.5 fm	0.27 ms
Stellar (e.g. Cyg X-1)	4×10^{34}	62 km	Peierls-locked
SMBH (e.g. Sgr A*)	8×10^{39}	1.2×10^{7} km	Peierls-locked

Table 2: Geometric evaporation lifetimes across the BH mass spectrum. Microscopic PBHs evaporate rapidly; macroscopic BHs are Peierls-locked and effectively stable on this channel.

6 Falsifiable Predictions

The SSM framework predicts four distinctive signatures, applicable across the full BH mass range:

Universal: no higher-level information recovery (no Page curve). The framework predicts that any black-hole evaporation—PBH, stellar-mass, or supermassive—reduces L2–L4 structure to the L1 thermal flux; higher-level information is not recoverable from the radiation. Observable consequences hold for all black holes:

- The Hawking radiation spectrum is exactly thermal at all times, with no Page-curve-style decrease in entanglement entropy past any time scale.
- Evaporation products are characterized entirely by the macroscopic conserved charges (mass, angular momentum, possibly $U(1)$ charge) of the initial BH; the L2–L4 quantum-state details of infalling matter do not survive as recoverable information in the radiation.
- Any future experimental or astrophysical demonstration of Page-curve information recovery would falsify the framework.

Macroscopic black holes: Hawking-only decay. For $R_H \gg L_{\text{corr}}$, Peierls locking suppresses the geometric channel to negligible levels (Table 2). Stellar-mass and supermassive black holes therefore evaporate exclusively through the standard Hawking channel, with lifetimes $\tau \propto M^3$ that exceed the age of the universe by many orders of magnitude. The framework is consistent with all existing astrophysical observations of black holes; no deviation from standard GR + Hawking phenomenology is expected for $R_H \gg 1$ fm.

PBH-specific: sharp mass cutoff at $\sim 10^{15}$ g. The $\tau \propto M^2$ scaling, combined with Peierls locking at $L_{\text{corr}} \sim 1$ fm, predicts that the present-day PBH mass spectrum should have a sharp lower cutoff near $M \sim 10^{15}$ g (where $R_H \sim L_{\text{corr}}$), with zero surviving relics below this threshold. This is distinct from the gradual roll-off predicted by Hawking $\tau \propto M^3$ scaling. Future PBH-search experiments sensitive to gamma-ray relic signals at sub-threshold masses would distinguish the two scaling laws and provide the cleanest observational test of the geometric channel.

Lorentz-invariant evaporation at leading order. The geometric channel obeys exact $SO(3)$ rotation isotropy at leading lattice order [1,2], with the first lattice anisotropies entering at $O((R_H/a)^{-4})$ via the degree-six fundamental invariant of the FCC point group [4]. For PBHs at $R_H \sim$ fm, this anisotropy is suppressed by a factor of $\sim (\ell_P/\text{fm})^4 \sim 10^{-80}$, undetectable in any plausible observation.

7 Discussion and Conclusion

The topological-vacancy framework reproduces all standard black-hole thermodynamic properties from the discrete CSS-stabilizer-code structure of the SSM vacuum (Section 2), derives a geometric evaporation channel from absolute quantum rate theory at the K = 12/K = 0 phase boundary (Section 4), and provides a hierarchical-reduction resolution of the BH information question (Section 3). These results hold for black holes of any mass. Their observational consequences split sharply at the QCD-set scale $L_{\text{corr}} \sim 1$ fm: for $R_H \gg L_{\text{corr}}$ (stellar-mass and supermassive BHs) the geometric channel is Peierls-locked (Section 5) and only Hawking decay survives, recovering standard astrophysical phenomenology; for $R_H \lesssim L_{\text{corr}}$ (PBHs near 10^{15} g) the geometric channel dominates, giving $\tau_{\text{Geo}} \propto M^2$ and resolving the long-standing PBH gamma-ray constraint.

The information question is addressed by recognizing that information in the discrete vacuum is hierarchical, with four levels (bond, plaquette, cuboctahedral cell, global stabilizer pattern). Across the K = 12 → K = 0 phase boundary the hierarchy itself does not survive: it reduces to the lowest level. Bond-level (L1) information is preserved as thermal Hawking radiation, reproducing the Bekenstein–Hawking entropy $S = A/(4G)$ as the L1 bit count; higher-level (L2–L4) structure—the combinatorial configuration of which Bell pairs occupied which edges, organized into stabilizer plaquettes and cuboctahedral cells—reduces to that thermal flux and is not recoverable as structure. Unitarity at the bond level is fundamental and globally preserved; unitarity at higher levels is an emergent property of the K = 12 phase that does not extend across phase boundaries. The standard information paradox dissolves: there is no contradiction because the strict L4 unitarity it assumes is an emergent property of the K = 12 phase, which a black-hole horizon by definition does not respect. The substantive position is: *total information is preserved at the lowest level, but the information hierarchy is not*—the same pattern that governs every irreversible thermodynamic reduction in nature.

Hawking's original 1975 conclusion [5] of exact-thermal radiation is recovered as the L1 channel of the framework. Page-curve recovery of higher-level structure [11] is explicitly predicted to be absent, providing a sharp empirical test.

Open questions: the detailed dynamics of L2–L4 reduction at the metric wall (this Letter treats it as an effective phase transition without specifying the microscopic decoherence path); the relation between the SSM phase-boundary picture and recent replica-wormhole derivations of the Page curve in toy models [12, 13], which we have argued lie outside the SSM regime of applicability but which deserve deeper comparison; and the cosmological-scale information content of the post-Big-Bang $\mathrm{K} = 12$ vacuum, whose L4 state is freshly crystallized through the $\mathrm{K} = 0 \to \mathrm{K} = 6 \to \mathrm{K} = 4 \to \mathrm{K} = 12$ sequence of the matter paper [1]. We address these in future work.

Acknowledgments. This work builds on the FCC lattice studies of [1–4] and the broader SSMTheory program.

Data availability. A Python script verifying all numerical claims of this paper (FCC geometry, metric wall, Bekenstein–Hawking entropy recovery, geometric lifetime formulas, PBH evaporation time, Peierls suppression scales) is available at `https://github.com/raghu91302/ssmtheory/blob/main/verify_pbh.py`.

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